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(Project-) Title: A Hand Movement Capture and Replay System for Subject Guidance in Motor Studies Using a Realistic 3D Hand Model and Camera Recordings

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1. Topic

Presenting effective and informative instructions to patients is very important during rehabilitation or EEG (electroencephalography) motor experiments that require patients to perform certain hand movements. Textual or image-based instructions alone are often insufficient for them to fully understand what movements they should do, which can negatively impact the experiment outcomes. As a result, accurate and realistic animations are a necessity for clearly demonstrating intended movements and ensuring consistent movements execution.

Modern game engines, such as Unreal Engine 5 (UE5), can be a solution because they provide a powerful environment for creating high quality animations through advanced rendering and rigging systems. However, producing animations in game engines typically requires specialized knowledge of animation pipelines (keyframing, control rigs, inverse kinematics, etc.), which poses a significant barrier for people with no experience with animations in game engines.

MediaPipe, an open-source framework from Google, can help address this gap by enabling us to extract human joint coordinates from each frame of a video. However, mapping these coordinates directly to a humanoid model in Unreal Engine leads to non-optimal visual results, such as noise, jittering and inconsistencies in proportion. Therefore, this thesis has 2 primary goals. First, it aims to provide a proper pipeline for producing high-quality hand instructions by turning a recorded motion video to a reusable animation in Unreal Engine without requiring the author to interact with Unreal's animation system directly. Second, it will improve the accuracy of MediaPipe's visual results by investigating several filtering strategies.

2. State of science and technology

Hand tracking and gesture recognition have been widely explored in recent years, one way is using specialized equipment such as leap motion devices [1] or smart gloves [2], which can provide high accuracy but impractical for widespread use due to their cost and setup complexity. Mediapipe, which also serves as one of the most adopted frameworks, can help to provide a more accessible solution to these limitations. However, the motion data from MediaPipe contains noise and jitter, motivating several works to focus on improving accuracy through filtering techniques. For instance, Zhang et al. [3] applied smoothing and Kalman filtering to MediaPipe outputs to stabilize landmarks for gesture recognition. Similarly, Li et al. [4] presented a comparable application but without detailing the filtering methods used. Other research has extended MediaPipe for more immersive experiences, such as the work of Kovács et al. [5], who introduced correction coefficients to enhance motion realism in interactive scenarios. Beyond MediaPipe, some studies have explored the use of multiple sensors combined with Extended Kalman Filters [6], highlighting the general importance of noise reduction and motion stabilization in hand-tracking systems.

MediaPipe has also been applied in real-time monitoring contexts. For example, Chen et al. [7] combined MediaPipe and OpenCV to implement real-time gesture recognition and exercise monitoring, demonstrating its potential in healthcare and rehabilitation. Similarly, Srecharan [8] experimented with MediaPipe-based real-time hand rigging, although the focus remained on gesture recognition rather than full-hand visualization accuracy.

A smaller set of works have attempted to integrate MediaPipe directly into game engines. The Mediapipe4U plugin [9] enables Unreal Engine users to leverage AI-driven motion capture but does not provide systematic details on filtering or scaling. Likewise, UnityHandTrackingWithMediapipe [10] demonstrates a working integration in Unity, but it lacks documentation on pipeline design, filtering, and inverse kinematics retargeting.

Additionally, studies also published several filtering evaluation metrics such as RMSE (Root Mean Square Error), MAE (Mean Absolute Error) [11], which assess motion smoothness by calculating the difference between the filtered data and ground truth data (the actual reference data). However, ground truth data is not always available, and in such cases, the filtering metrics can be evaluated using MDCSS (Movement Dynamics Clutter Spectrum Strength) to detect jitter by measuring the consistency of velocity direction[12].

In summary, most published works concentrate on movement recognition and do not explicitly describe a proper setup for real-time hand tracking in Unreal Engine with appropriate filtering, scaling, and retargeting strategies. This leaves a gap in the literature for research focused on stabilizing and optimizing MediaPipe data for realistic visualization in advanced 3D environments such as Unreal Engine, which is one of the goals of this thesis.

Furthermore, this project will also adopt Unreal Engine 5 as the target environment for generating replayable hand movement via camera recordings. Unreal Engine 5 has been popularly known for its realistic visualization, thanks to their advanced lighting, material, physics visualization, making it particularly suitable for producing natural-looking and realistic hand movements. In addition, UE5 also provides us with a realistic human model called Metahuman, improving realism and visual quality.

3. Objective of your project

This thesis aims to develop a complete end to end pipeline that allows people with no prior knowledge of game engine animation systems to produce high-quality hand gesture instruction animations from a recorded motion video. To achieve this, there are several steps that need to be processed:

1. Establish MediaPipe Data Extraction
   Write a python script to capture videos using standard webcam, followed by extracting the 21 landmarks data per frame using MediaPipe Hands API (docs). The robustness of MediaPipe data also can be improved by adjusting the minimum tracking confidence score, which is a threshold between 0.0 and 1.0 that determines

whether MediaPipe considers the hand tracking between consecutive frames reliable.

MediaPipe detects 21 landmarks per hand, consisting of one landmark on the wrist, which serves as a base of the hand's hierarchy, where it will act as a reference point for all other finger joints. Thumb, index finger, middle finger, ring finger, pinky finger have 4 landmarks each.

2. Data Filtering
   The hand-tracking data of MediaPipe cannot be directly integrated into Unreal Engine 5 without additional processing. Therefore, it is necessary to investigate several filtering methods such as standard linear Kalman filter and One Euro filter, which are widely used for motion smoothing, to improve stability of the resulting motion. The evaluation of these filtering methods also plays an important role, which can be conducted by calculating MDCSS, as reported in the state of science and technology section.

3. Implement Rigging System for Humanoid Model
   The framework first requires setting up a control rig in UE5, which serves as the foundation for evaluating Inverse Kinematic (IK) solutions. UE5 provides multiple IK solvers, such as TwoBoneIK, Full Body IK (FBIK), FABRIK, and can be combined with joint constraints to produce more realistic motions.
   TwoBoneIK: IK solver for three-joint chains, consisting of root joint, intermediate joint, end effector, it computes joint rotations using triangle geometry.
   Full Body IK: solving IK across the entire selected skeleton parts simultaneously and optimizing it to satisfy all constraints.
   FABRIK: Forward and Backward Reaching Inverse Kinematics, works iteratively by manipulating joint positions directly, rather than joint angles.

4. Integrate All Components into Pipeline and Documentation
   The complete end-to-end pipeline integrates MediaPipe data extraction and filtering MediaPipe data using Python script. The processed landmarks data will be exported to UE5 in CSV format, enabling realistic hand movement animations with a proper IK setup. Documenting edge cases where pipeline fails for example gesture too fast, unsuitable lighting condition, hand partially hidden, will also be helpful.

4. Work plan, necessary resources

Week 1: Conduct a literature review on hand tracking, MediaPipe, inverse kinematics, filtering methods

Week 2: Develop a Python script to generate raw hand tracking data from MediaPipe

Week 3 - 4: Set up the UE5 environment, including Control Rig configuration on a humanoid model and integration of MediaPipe data into UE5.

Week 5: Visualize raw MediaPipe data in UE5 to establish a baseline for comparison with the final animation results.

Week 5 - 7: Applying Filtering Methods (Standard Linear Kalman Filter and One Euro Filter) and visualizing the results of each approach.

Week 8: Define and compute evaluation metrics for filtering performance.

Week 9 - 10: Perform qualitative analysis and comparative evaluation of the results.

Week 10 - 11: Integrate all components into one pipeline and perform testing.

Week 11 - 12: Finalize project, complete the paper and prepare paper defense.

Gantt Chart : Gantt Chart Bachelor Thesis

5. Literature

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6. Structure

<u>Overview</u>

This thesis covers the intersection of hand motion capture, data processing, and real-time 3D animation for instructional and rehabilitation purposes. The main topics include:

1. Patient Instruction and Rehabilitation Visualization
   The importance of clear, accurate hand movement instructions in rehabilitation and EEG motor experiments, and the limitations of text- or image-based guidance.

2. Vision-Based Hand Tracking with MediaPipe
   Use of MediaPipe Hands to extract 21 hand landmarks per frame from monocular video, focusing on accessibility and low-cost motion capture.

3. Noise, Jitter, and Data Quality in MediaPipe
   Analysis of typical artifacts in MediaPipe output and their impact on visual realism when mapped to 3D humanoid models.

4. Inverse Kinematics and Rigging in Unreal Engine 5
   Application of Unreal Engine 5 control rigs and IK solvers (TwoBoneIK, FABRIK, Full Body IK) to retarget filtered landmark data onto a realistic humanoid hand model.

5. Filtering and Motion Stabilization Techniques
   Investigation of temporal filtering methods (Kalman filter, One Euro filter) to improve motion smoothness and stability, along with qualitative evaluation metrics (e.g., MDCSS).

6. End-to-End Animation Pipeline Design
   Design of a reproducible, reusable pipeline that transforms recorded video into replayable, high-quality hand gesture animations without requiring direct interaction with UE5’s animation system.

<u>Focus and guiding thread of the work</u>

This work focuses on developing an end-to-end, user-friendly pipeline that converts webcam-recorded hand movements into stable, realistic, and reusable hand-gesture animations in Unreal Engine 5 by filtering and retargeting MediaPipe data without requiring animation expertise.

Content presentation created by: München, 16/01/2026	Garching, 15/01/2026	
Place, date	*Place, date*	*Place, date*
Student	First examiner (supervisor)	Second examiner (Center of Key Competenci